Superconductivity: Fundamental Principles and Key Effects

Review Report

Introduction

Superconductivity was discovered by Heike Kamerlingh Onnes in 1911. It is one of the most interesting quantum effects in physics. When some materials are cooled below a critical temperature T_c, they change into a special state with two unique properties: perfect electrical conduction and complete expulsion of magnetic fields. This review explains the basic physics of superconductivity by focusing on three main effects.

The Meissner Effect: Magnetic Field Expulsion

The Meissner effect is not just a result of zero resistance. It is a separate fundamental property. Below T_c, the material becomes a perfect diamagnet and pushes out all magnetic fields from inside.

A special relationship between current and magnetic field exists in superconductors — the first London equation:

$$\nabla \times \mathbf{J}_s = -\frac{1}{\mu_0 \lambda_L^2} \mathbf{B}$$

where $\mathbf{J}_s$ is the supercurrent density, λ_L is the characteristic length.

Using Maxwell's equation for steady state:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}_s$$

Taking the curl of this equation and using the first London equation, we get:

$$\nabla^2 \mathbf{B} = \frac{1}{\lambda_L^2} \mathbf{B}$$

The solution for a semi-infinite superconductor ($x \geq 0$):

$$B(x) = B_0 e^{-x/\lambda_L}$$

where $\lambda_L = \sqrt{\frac{m}{\mu_0 n_s (2e)^2}}$ is the London penetration depth.

Physical Interpretation:

- Magnetic field enters the superconductor only up to depth λ_L (~100 nm)
- Surface currents create shielding that cancels external field inside
- These currents flow without energy loss due to superconductivity
- The effect works only below critical temperature T_c and field H_c

Significance:

The Meissner effect shows that superconductivity is a true thermodynamic phase, not just zero resistance. The complete magnetic field expulsion is a key sign of the superconducting transition and is used in many applications.

Flux Quantization: Macroscopic Quantum Effect

Superconductors show clear macroscopic quantum behavior through magnetic flux quantization. In a superconducting ring, the trapped magnetic flux is quantized in units of:

$$\Phi = n\Phi_0, \quad \Phi_0 = \frac{h}{2e} = 2.07 \times 10^{-15} \text{ Wb} \tag{1}$$

This happens because the superconducting wavefunction must be single-valued. We can write the wavefunction as $\psi(\mathbf{r}) = \sqrt{n_s}e^{i\theta(\mathbf{r})}$, where θ is the phase. For a closed loop around the flux, the phase must change by integer multiples of 2π:

$$\oint \nabla\theta \cdot d\mathbf{l} = 2\pi n \tag{2}$$

Using the momentum expression $\mathbf{p} = \hbar\nabla\theta - 2e\mathbf{A}$, we get the flux quantization condition. The $2e$ factor shows that current is carried by electron pairs, not single electrons.

Josephson Effects: Quantum Phase Effects

Brian Josephson predicted in 1962 that Cooper pairs can tunnel between two superconductors separated by a thin insulator. The Josephson effects show that the superconducting phase has real physical effects.

The DC Josephson effect gives supercurrent without voltage:

$$I = I_c \sin(\varphi) \tag{3}$$

where $\varphi = \theta_1 - \theta_2$ is the phase difference, and I_c is the critical current.

With DC voltage V, the AC Josephson effect occurs:

$$\frac{d\varphi}{dt} = \frac{2eV}{\hbar} \tag{4}$$

This creates an oscillating supercurrent with frequency $\omega_J = \frac{2eV}{\hbar}$. This frequency-voltage relation is exact and is used for precise measurements of e/h.

Conclusion

The Meissner effect, BCS theory, flux quantization, and Josephson effects together show superconductivity as a macroscopic quantum phenomenon. These principles explain the special properties of superconductors and enable important technologies like MRI machines, particle accelerators, and quantum computers. While high-temperature superconductors still present challenges, the basic quantum principles of conventional superconductors remain fundamental to our understanding.